

Numerical Solutions of Laplace Equation

Comparison of Steady-State and Transient Methods

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Computational Physics II

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Outline

- 1 Problem Definition
- 2 Numerical Methods
- 3 Numerical Formulation
- 4 Results



Problem Definition

Solve $\nabla^2 \psi(x, y) = 0$ on $[0, 3] \times [0, 3]$,
subject to:

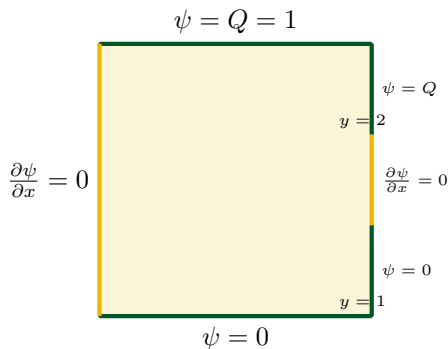
$$\psi(x, 3) = Q = 1, \quad 0 \leq x \leq 3,$$

$$\psi(3, y) = \begin{cases} 1, & 2 \leq y \leq 3, \\ 0, & 0 \leq y \leq 1, \end{cases}$$

$$\psi(x, 0) = 0, \quad 0 \leq x \leq 3,$$

$$\frac{\partial \psi}{\partial x}(0, y) = 0, \quad 0 \leq y \leq 3,$$

$$\frac{\partial \psi}{\partial x}(3, y) = 0, \quad 1 \leq y \leq 2.$$



Steady-State Methods

- **SOR**

- $\omega^* = \frac{2}{1 + \sin(\pi/(N-1))}$

- **Conjugate Gradient**

- solves the sparse SPD system
 - BCs embedded in the RHS

Transient Methods

- **Forward Euler**

- CFL: $\Delta t \leq \frac{h^2}{4}$
 - first-order in time

- **Runge-Kutta 4**

- fourth-order in time
 - larger stable step



Numerical Formulation

Finite-Difference Approximation

- Five-point stencil

$$-4\psi_{i,j} + \psi_{i+1,j} + \psi_{i-1,j} + \psi_{i,j+1} + \psi_{i,j-1} = 0 \quad A\Psi = b, \quad A: \text{ sparse, SPD}$$

- Neumann (2nd-order ghost)

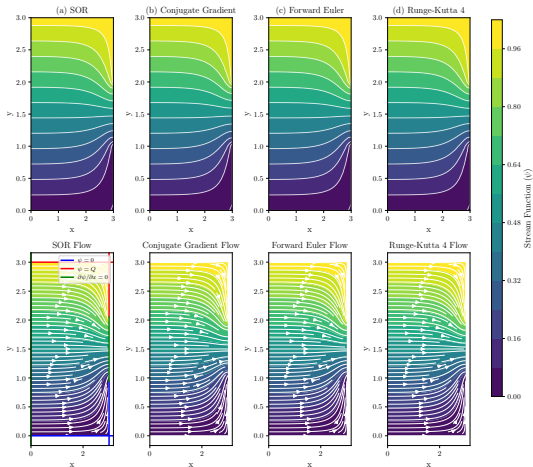
$$\psi_{0,j} = \frac{4}{3} \psi_{1,j} - \frac{1}{3} \psi_{2,j}$$

Matrix Formulation (CG)

- $-4/h^2$ on the diagonal
- $+1/h^2$ for each neighbour
- Extra $+1/h^2$ on diag at Neumann faces
- Dirichlet / ghost terms $\rightarrow b$

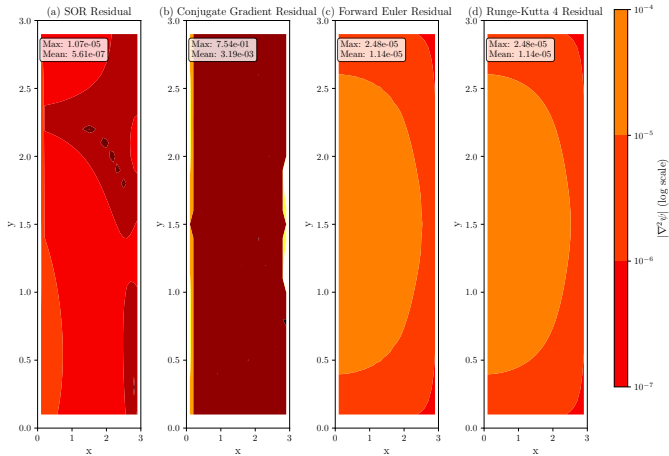
Numerical Methods – default case

Comparison of Numerical Methods for Laplace Equation
Grid: $h=0.1$, Boundary: $Q=1.0$, Neumann: default



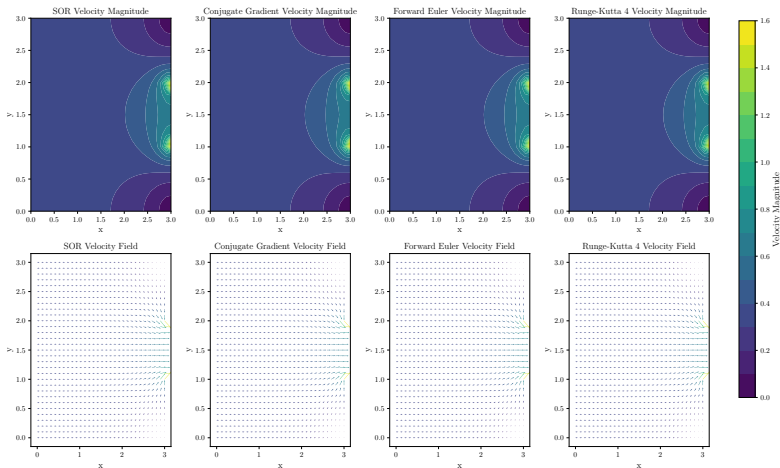
Error Map ($h = 0.1$)

Laplacian Residual Comparison
Grid: $h=0.1$, Boundary: $Q=1.0$, Neumann: default



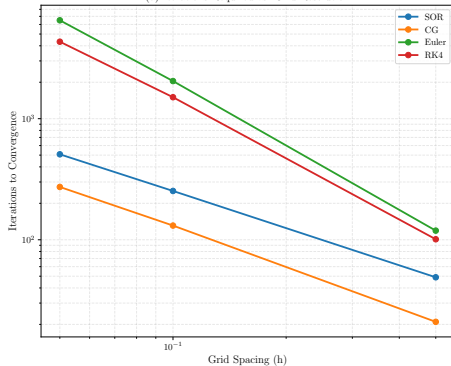
Velocity Field ($h = 0.1$)

Velocity Field Comparison
Grid: $h=0.1$, Boundary: $Q=1.0$, Neumann: default

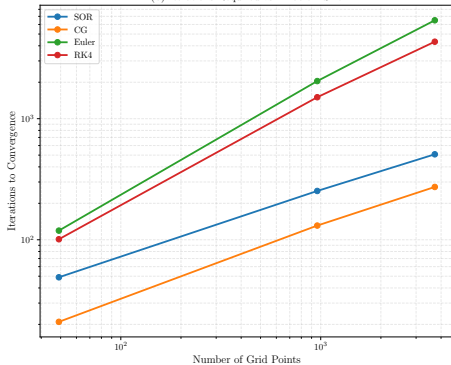


Iterations vs Problem Size

(a) Iterations Required vs. Grid Resolution



(b) Iterations Required vs. Problem Size



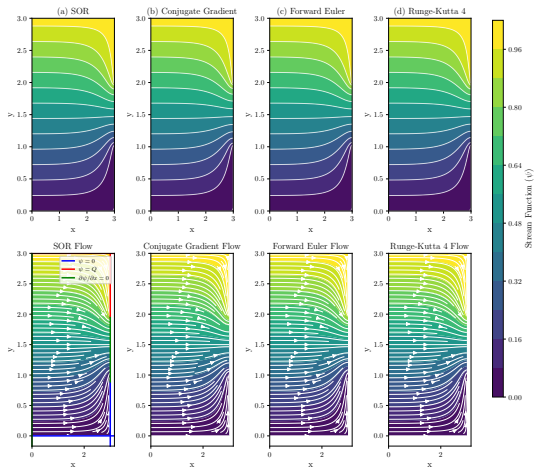
Conclusions

- All four solvers reach the **same steady field** for every boundary mix (Dirichlet, Neumann, or mixed).
- **Who is fastest?**
 $\text{CG} < \text{SOR} (\omega \approx \omega_{\text{opt}}) \ll \text{RK4} < \text{Euler}.$
- All retain the intended $O(h^2)$ **spatial accuracy**; second-order ghost points keep Neumann walls consistent.
- **Take-away:** use CG for production speed, **but** relatively harder to code in.



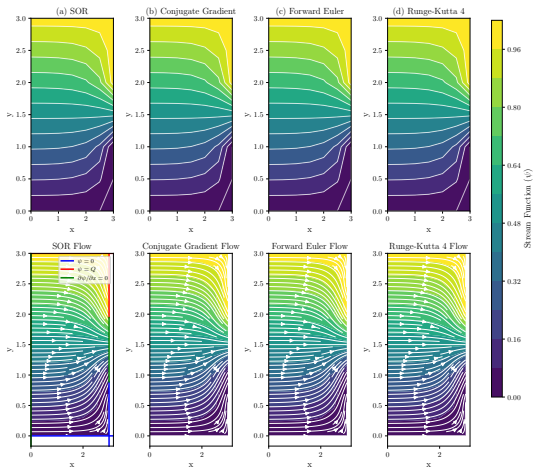
Method Comparison — Fine Grid ($h = 0.05$)

Comparison of Numerical Methods for Laplace Equation
Grid: $h=0.05$, Boundary: $Q=1.0$, Neumann: default



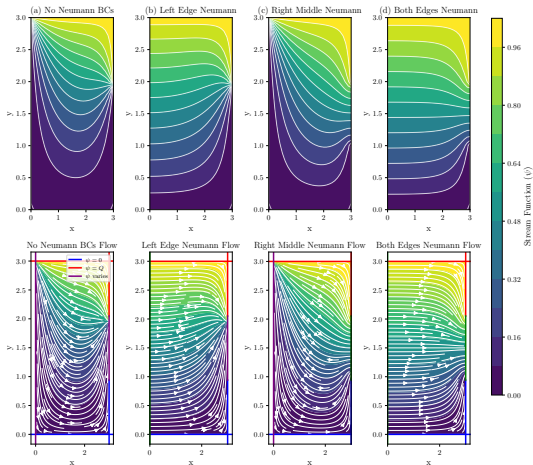
Method Comparison — Coarse Grid ($h = 0.5$)

Comparison of Numerical Methods for Laplace Equation
Grid: $h=0.5$, Boundary: $Q=1.0$, Neumann: default



Effect of Neumann-Boundary Placement (SOR)

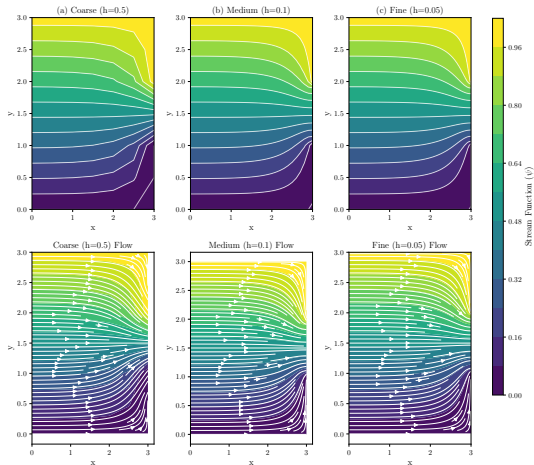
Effect of Different Neumann Boundary Conditions (SOR Method)
Grid: $h=0.1$, Boundary: $Q=1.0$



Grid-Refinement Study (SOR)

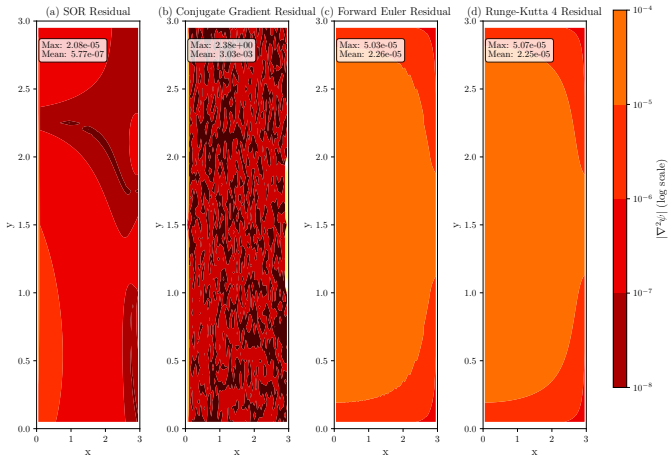
Effect of Grid Refinement (SOR Method)

Boundary: $Q=1.0$, Neumann: default



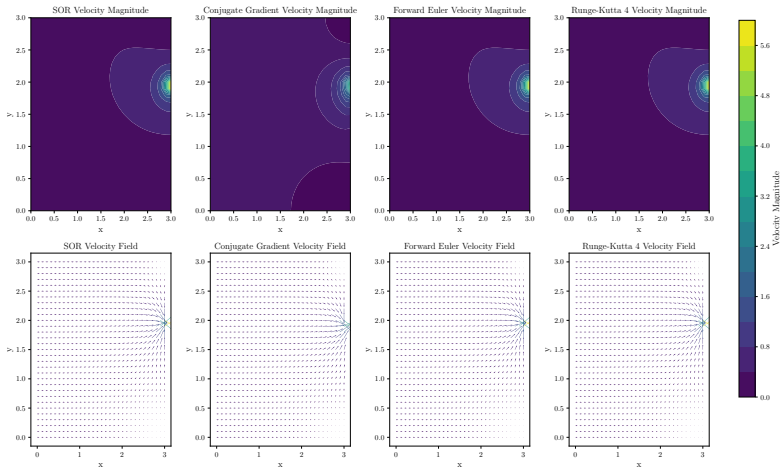
Laplacian Residual Map — Fine Grid

Laplacian Residual Comparison
Grid: $h=0.05$, Boundary: $Q=1.0$, Neumann: default



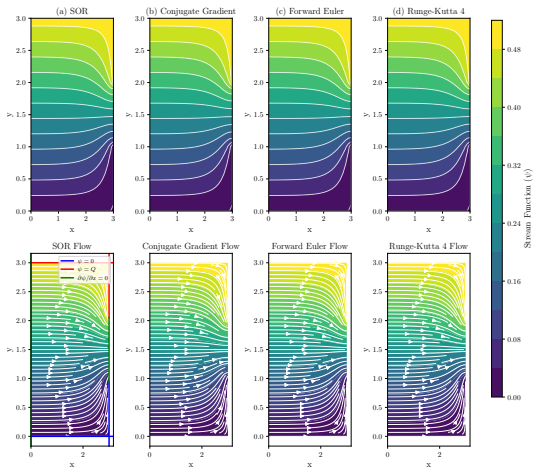
Velocity Field — Left-Edge Neumann Only

Velocity Field Comparison
Grid: $h=0.1$, Boundary: $Q=1.0$, Neumann: left_only



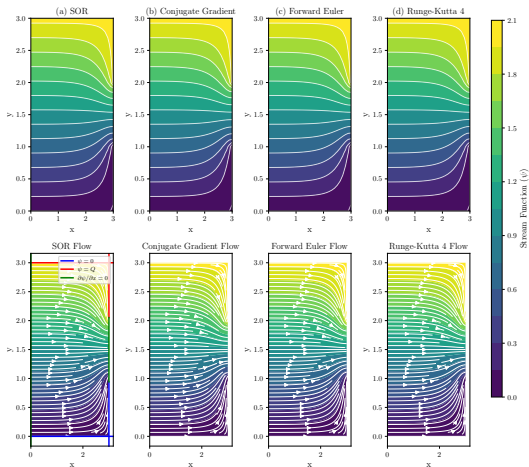
Method Comparison — Boundary Value $Q = 0.5$

Comparison of Numerical Methods for Laplace Equation
Grid: $h=0.1$, Boundary: $Q=0.5$, Neumann: default



Method Comparison — Boundary Value $Q = 2.0$

Comparison of Numerical Methods for Laplace Equation
Grid: $h=0.1$, Boundary: $Q=2.0$, Neumann: default



Thank You!

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